Trace a campus connection

Identify a Student access port, its VLAN and its path to the default gateway.

EXCHANGE TO INSPECT

PC0	SW4-STUDENTS	SW1-CORE	R1-CAMPUS

1 Locate the endpoint

Open Topology guide, select Students, then choose Locate PC0. In Packet Tracer, open PC0 > Desktop > Command Prompt.

```
ipconfig /all
```

Expect a /26 lease in 192.168.10.2-62, gateway 192.168.10.1 and DNS 192.168.10.130. Lease order can change.

2 Inspect the access link

On SW4, open CLI in privileged EXEC mode. Find PC0 on Fa0/1. Use the inventory or link inspector to confirm the far end.

```
show vlan brief
```

Fa0/1 belongs to VLAN 30. An endpoint sends ordinary Ethernet frames on this access link.

3 Inspect both trunks

On SW4 and SW1, inspect operational trunks and allowed VLANs. Follow the uplink toward R1.

```
show interfaces trunk
```

SW4 Gi0/1 to SW1 Fa0/24 carries VLANs 10 and 30. SW1 Gi0/1 to R1 Gi0/0 carries all four VLANs.

4 Verify the gateway

On R1, inspect the IP interfaces. From PC0, test the Student gateway and compare with Lab results A02.

```
show ip interface brief | PC0: ping 192.168.10.1
```

R1 terminates VLAN 30 at Gi0/0.30. An echo reply confirms the client can reach its gateway.

TROUBLESHOOTING

If PC0 shows 0.0.0.0, allow about 60 seconds for convergence, then renew DHCP. Check the access VLAN and trunk allowance before changing any address.